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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2025

Fourth Semester

21CSC204J - DESIGN AND ANALYSIS OF ALGORITHMS

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Marks CO BL

Answer all Questions

- | | | | |
|--|---|---|---|
| 1. The best-case time complexity of Insertion Sort is typically represented by which asymptotic notation? | 1 | 1 | 1 |
| <div style="display: flex; justify-content: space-between;"> <div>(A) $O(n^2)$</div> <div>(B) $O(n \log n)$</div> </div> <div style="display: flex; justify-content: space-between;"> <div>(C) $O(n)$</div> <div>(D) $O(1)$</div> </div> | | | |
| 2. When analyzing the time complexity of an algorithm, the "operation count" refers to: | 1 | 1 | 1 |
| <div style="display: flex; justify-content: space-between;"> <div>(A) The number of lines of code in the algorithm.</div> <div>(B) The total number of instructions executed by the CPU.</div> </div> <div style="display: flex; justify-content: space-between;"> <div>(C) The number of variables declared in the algorithm.</div> <div>(D) The number of times the algorithm is run.</div> </div> | | | |
| 3. Which method for solving recurrence relations involves making an educated guess for the solution and then using mathematical induction to prove its correctness? | 1 | 1 | 2 |
| <div style="display: flex; justify-content: space-between;"> <div>(A) Recursion Tree Method</div> <div>(B) Master Theorem</div> </div> <div style="display: flex; justify-content: space-between;"> <div>(C) Substitution Method</div> <div>(D) Iteration Method</div> </div> | | | |
| 4. If an algorithm's running time is described as $\Omega(n^2)$, what does this notation signify? | 1 | 1 | 2 |
| <div style="display: flex; justify-content: space-between;"> <div>(A) The algorithm's running time is exactly n^2.</div> <div>(B) The algorithm's running time is at most n^2.</div> </div> <div style="display: flex; justify-content: space-between;"> <div>(C) The algorithm's running time is approximately n^2.</div> <div>(D) The algorithm's running time is at least n^2.</div> </div> | | | |
| 5. Consider Strassen's Matrix multiplication algorithm. If the standard matrix multiplication has a time complexity of $O(N^3)$, and Strassen's reduces this, which of the following recurrence relations accurately represents the time complexity of Strassen's algorithm for multiplying two $N \times N$ matrices? | 1 | 2 | 1 |
| <div style="display: flex; justify-content: space-between;"> <div>(A) $T(N) = 8T(N/2) + O(N^2)$</div> <div>(B) $T(N) = 8T(N/2) + O(N \log N)$</div> </div> <div style="display: flex; justify-content: space-between;"> <div>(C) $T(N) = 7T(N/2) + O(N^3)$</div> <div>(D) $T(N) = 7T(N/2) + O(N^2)$</div> </div> | | | |
| 6. An algorithm for finding the largest sub-array sum using a divide-and-conquer approach has a recurrence relation of $T(N) = 2T(N/2) + O(N)$. According to the Master Theorem, what is the time complexity of this algorithm? | 1 | 2 | 3 |
| <div style="display: flex; justify-content: space-between;"> <div>(A) $O(N \log N)$</div> <div>(B) $O(N)$</div> </div> <div style="display: flex; justify-content: space-between;"> <div>(C) $O(N^2)$</div> <div>(D) $O(\log N)$</div> </div> | | | |

7. Choose any one from the following statements about the worst-case time complexity of Quick Sort is TRUE. 1 2 2
- (A) It is $O(N \log N)$ when the pivot selection consistently divides the array into roughly equal halves. (B) It is $O(N^2)$ when the pivot selection consistently leads to highly unbalanced partitions
- (C) It is $O(N)$ when the input array is already sorted in ascending or descending order. (D) It is always $O(N \log N)$ regardless of pivot selection.
8. Consider an algorithm to find both the maximum and minimum elements in an array of N elements. A naive approach would involve two separate linear scans, each taking $O(N)$ time. What is the minimum number of comparisons (in terms of N) required in the worst case to find both the maximum and minimum elements using a more optimized approach? 1 2 2
- (A) $2N-2$ (B) $N-1$
(C) $3N/2-2$ (D) $N \log N$
9. Identify which of the following statements regarding the 0/1 Knapsack problem and the Fractional Knapsack problem is TRUE. 1 3 2
- (A) Both 0/1 Knapsack and Fractional Knapsack can be optimally solved using a greedy approach. (B) 0/1 Knapsack requires dynamic programming for an optimal solution, while Fractional Knapsack can be solved greedily.
- (C) Fractional Knapsack requires dynamic programming for an optimal solution, while 0/1 Knapsack can be solved greedily. (D) Neither 0/1 Knapsack nor Fractional Knapsack can be optimally solved using a greedy approach.
10. In a scenario where a Huffman tree is being constructed for a set of characters with frequencies A:5, B:9, C:12, D:13, E:16, and F:45, which pair of characters would be the first to be combined into a new node during the Huffman coding process? 1 3 2
- (A) A and B (B) A and E
(C) C and D (D) B and C
11. When comparing Kruskal's and Prim's algorithms for finding a Minimum Spanning Tree (MST), which of the following is a key distinguishing factor in their greedy strategies? 1 3 2
- (A) Kruskal's algorithm adds the cheapest edge that connects two previously unconnected components, while Prim's algorithm adds the cheapest edge connecting a vertex in the MST to a vertex outside the MST. (B) Kruskal's algorithm always starts from a specific source vertex, while Prim's algorithm does not.
- (C) Prim's algorithm uses a Disjoint Set Union data structure, while Kruskal's algorithm uses a priority queue. (D) Both algorithms produce different MSTs for the same graph.
12. Given two sequences, $X = \text{"AGGTAB"}$ and $Y = \text{"GXTXAYB"}$, what is the length of their Longest Common Subsequence (LCS)? 1 3 2
- (A) 6 (B) 5
(C) 4 (D) 3

13. Which of the following statements most accurately describes the relationship between Branch and Bound algorithms and Backtracking algorithms when solving the 0/1 Knapsack problem? 1 4 1
- (A) Branch and Bound is primarily used for maximization problems, while Backtracking is exclusively for decision problems.
 - (B) Backtracking is a specific type of Branch and Bound, where the bounding function is implicitly defined by the feasibility check of the current state.
 - (C) Both Branch and Bound and Backtracking are identical in their approach to solving optimization problems like the 0/1 Knapsack problem.
 - (D) Branch and Bound is a specific type of Backtracking that uses bounding functions to prune the search space, while Backtracking explores all possible solutions.
14. Consider a graph with N vertices. What is the time complexity of finding all-pairs shortest paths using the Floyd-Warshall algorithm, and how does it compare to running Dijkstra's algorithm from each vertex on a dense graph represented by an adjacency matrix? 1 4 2
- (A) Floyd-Warshall is $O(N^3)$; Dijkstra from each vertex is $O(N^3)$ for dense graphs.
 - (B) Floyd-Warshall is $O(N^2)$; Dijkstra from each vertex is $O(N^2 \log N)$ for dense graphs.
 - (C) Floyd-Warshall is $O(N^3)$; Dijkstra from each vertex is $O(N^4)$ for dense graphs.
 - (D) Floyd-Warshall is $O(N^3)$; Dijkstra from each vertex is $O(N^2 \log N)$ for dense graphs.
15. When applying Branch and Bound algorithms to the Traveling Salesman Problem (TSP), what is the primary role of the bounding function? 1 4 1
- (A) To determine the exact cost of the optimal tour at each node of the state-space tree.
 - (B) To provide an upper bound on the cost of any tour that can be formed from the current partial solution.
 - (C) To provide a lower bound on the cost of any tour that can be formed from the current partial solution, allowing for pruning of subtrees.
 - (D) To identify if a Hamiltonian circuit exists from the current partial path.
16. In the context of the N-Queens problem solved using backtracking strategy, what is the significance of the PLACE(k) function? 1 4 1
- (A) It checks if a queen placed in row k can attack any queen in a previous row.
 - (B) It checks if a queen can be placed in column k without conflict.
 - (C) It determines the optimal position for the k -th queen to minimize conflicts.
 - (D) It calculates the total number of safe positions available for the k -th queen.
17. In the context of the Randomized Hiring Problem, what is the primary purpose of randomization? 1 5 1
- (A) To minimize the total number of candidates interviewed.
 - (B) To ensure the best candidate is always hired.
 - (C) To reduce the expected cost of hiring.
 - (D) To eliminate the need for interviews.

18. The Rabin-Karp algorithm primarily uses which technique to efficiently compare substrings with a pattern? 1 5 1
 (A) Dynamic Programming (B) Hashing
 (C) Greedy Approach (D) Divide and Conquer
19. For the Vertex Cover problem, an approximation algorithm aims to find a vertex cover whose size is: 1 5 2
 (A) Exactly equal to the optimal vertex cover size. (B) At most twice the size of the optimal vertex cover.
 (C) Always half the size of the optimal vertex cover. (D) Independent of the optimal vertex cover size.
20. An NP-Hard problem is characterized by which of the following properties? 1 5 1
 (A) Any NP problem can be reduced to polynomial time. (B) It can be verified in polynomial time.
 (C) It can be solved in polynomial time. (D) It is always easier to solve than an NP-Complete problem.

PART - B (5 × 8 = 40 Marks)

Marks CO BL

Answer all Questions

21. a) Illustrate the concept of time complexity analysis, elaborating on best, worst, and average-case scenarios using Insertion Sort as an example. How do asymptotic notations like O , Ω , and Θ relate to these analyses? 8 1 3

(OR)

- b) Draw the recursion tree for $T(n) = 4T(n/2) + cn$, where c is a constant, and provide a tight asymptotic bound on its solution. Verify the given bound value by the substitution method. 8 1 3

22. a) Apply mergesort to sort the list S, O, R, T, I, N, G in alphabetical order. Justify whether mergesort is a stable sorting algorithm or not. Solve the recurrence relation for the number of key comparisons made by mergesort in the worst case. It is assumed that $n = 2^k$. 8 2 4

(OR)

- b) Explain in detail the divide-and-conquer algorithm for solving the closest-pair problem in a set of points on a 2D plane. Then, prove that during the merge step, for every point p in the vertical strip of width $2d_{\min}$, the algorithm needs to examine no more than seven other points that could potentially be closer to p than $d_{\min}$, the minimum distance found so far. 8 2 3

23. a) Given two strings, $s1 = (1, 0, 0, 1, 0, 1, 0, 1)$ and $s2 = (0, 1, 0, 1, 1, 0, 1, 1, 0)$, devise a dynamic programming algorithm to find the length of their Longest Common Subsequence (LCS). The algorithm should utilize a 2D table to store and build upon the solutions of subproblems, ultimately determining the maximum length of a subsequence common to both $s1$ and $s2$. Additionally, describe how the 2D table is constructed and how the final LCS length is derived from it. 8 3 4

(OR)

- b) Elaborate how Huffman coding exemplifies both the greedy choice property and the optimal substructure property, which are fundamental characteristics of greedy algorithms. What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers?
 a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21 8 3 4

24. a) Apply backtracking method to solve the following instance of the subset sum problem: $A = \{1, 3, 4, 5\}$ and $d = 11$. What are the two inequalities used to terminate a node as nonpromising and what is its significance? 8 4 3

(OR)

- b) Trace the working of Floyd-Warshall algorithm on the weighted, directed graph; given by the matrix which shows the weighted directed edges between the graph. Show the matrix $D^{(k)}$ that results for each iteration of the outer loop. 8 4 3

Vertices	1	2	3	4	5	6
1					-1	
2	1			2		
3		2				-8
4	-4				3	
5		7				
6		5	10			

25. a) Explain how the Rabin-Karp string matching algorithm works, and analyse its preprocessing and runtime complexities? When the working modulo $q = 11$, how many spurious hits does the Rabin-Karp matcher encounter in the text $T = 3141592653589793$ when looking for the pattern $P = 26$? 8 5 4

(OR)

- b) Given a Boolean formula in conjunctive normal form (CNF), where each clause contains exactly three literals, determine whether the formula is satisfiable. Explain the structure of a 3-SAT instance, and discuss why 3-SAT is considered NP-complete. How does the complexity of solving 3-SAT compare to that of 2-SAT and general SAT problems? 8 5 4

PART - C ($1 \times 15 = 15$ Marks)

Marks CO BL

Answer any 1 Question

26. Given a collection of n bolts of different widths and n corresponding nuts, one is allowed to try a nut and a bolt together to determine whether the nut is larger than the bolt, smaller than the bolt, or matches the bolt exactly. However, it is not possible to compare two nuts directly or two bolts directly. The task is to design a Quicksort based algorithm that matches each bolt with its corresponding nut, achieving an average-case efficiency of $O(n \log n)$. 15 2 4

27. Determine the cost and structure of an optimal binary search tree for a set of keys with the following probabilities: 15 3 4

i	0	1	2	3	4	5	6	7
p_i		0.04	0.06	0.08	0.02	0.10	0.12	0.14
q_i	0.06	0.06	0.06	0.06	0.05	0.05	0.05	0.05

Give the recurrence relations used to fill the e-table (Expected Cost Table), w-table (Weight Table) and root-table (Root Index Table); also analyse the algorithm and determine its running time.

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